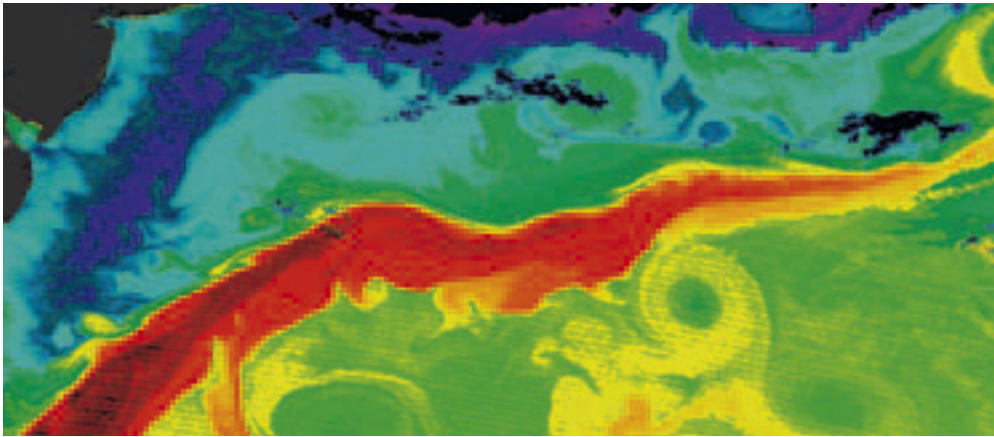




◀ OCEAN CURRENT DISRUPTION

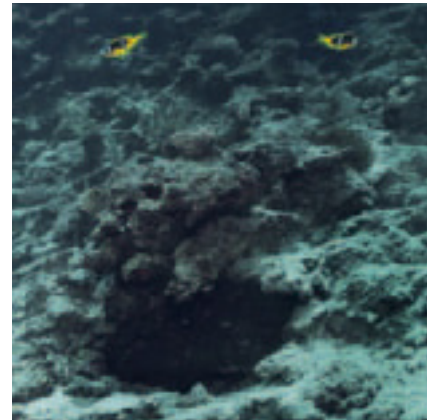
Fresh meltwater draining off melting continental ice sheets is diluting the water of nearby oceans, making it less salty, less dense, and less likely to sink.

This is weakening the sinking currents in the north Atlantic Ocean that help drive the warm Gulf Stream current, seen in red in this satellite image. If the Gulf Stream is disrupted, it will seriously affect the climate of northern Europe, and possibly the world.

climate
change

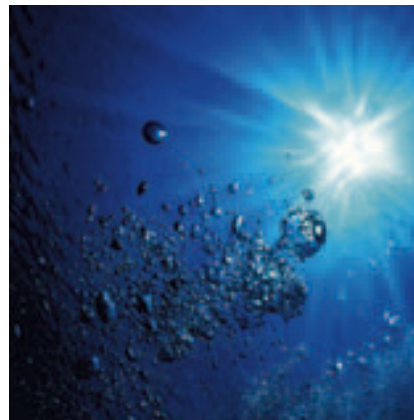
▶ CORAL BLEACHING ▶

Tropical reef corals rely on tiny organisms living in their tissues to provide them with food. These organisms cannot tolerate very warm water. It makes them leave the coral, which then turns white and may die. This coral bleaching damaged a quarter of the world's coral in 1998, and it could become an annual event, gradually destroying one of the world's richest habitats.



◀ METHANE THREAT

Carbon dioxide is not the only greenhouse gas causing climate change. It is also caused by methane gas, which is released by many natural processes. There are vast reserves of frozen methane buried in the sediments of the ocean floor. If rising ocean temperatures allow these to melt, they could bubble up through the water and into the atmosphere, dramatically accelerating global warming.



ACID OCEANS



A lot of the carbon dioxide that we have added to the atmosphere has dissolved in the oceans, but has made their waters more acidic. This acid attacks the chalky shells of crustaceans and mollusks such as crabs and clams, and destroys the limestone skeletons of corals. If the problem worsens it could kill off coral reefs worldwide, and cause mass extinction of oceanic shelled animals.

▲ STORM WARNING

Destructive hurricanes are fueled by the evaporation of warm ocean water. The warmer the oceans, the more fuel these storms have available, so warmer oceans may cause more storms. These could be more violent where ocean temperatures are highest, and they may also be more widespread. It may be significant that 2005 was the worst Atlantic hurricane season on record.

